| Dataset | Data subset | Non-white | White | Men | Women | Christian | Non-Christian | LGBT | Straight | Disability |
|---|---|---|---|---|---|---|---|---|---|---|
| Jigsaw Kaggle | - | 22445 | 27262 | 48629 | 58274 | 44485 | 31332 | 12063 | 1428 | 5522 |
| Jigsaw Bias | - | 19682 | 3028 | 1514 | 1514 | 4542 | 7570 | 15140 | 3028 | 4542 |
| Stormfront | - | 476 | 255 | 648 | 536 | 118 | 327 | 91 | 1 | 18 |
| TweetEval | Hate | 932 | 135 | 1956 | 4535 | 121 | 332 | 90 | 1 | 27 |
| TweetEval | Offensive | 303 | 58 | 779 | 780 | 101 | 84 | 73 | 2 | 56 |
| OpenAI | - | 184 | 46 | 384 | 400 | 38 | 101 | 117 | 5 | 32 |
| Movie Plots | - | 1000 | 239 | 330 | 2417 | 213 | 183 | 351 | 383 | 439 |
| TV Synopses | Short TMDB | 45 | 20 | 20 | 110 | 338 | 115 | 584 | 332 | 114 |
| TV Synopses | Medium Wiki | 60 | 10 | 20 | 50 | 188 | 53 | 324 | 189 | 74 |
| TV Synopses | Long IMDB | 26 | 11 | 30 | 137 | 70 | 17 | 207 | 94 | 34 |
| Traditional | - | 43838 | 30738 | 53526 | 65639 | 49367 | 39645 | 27457 | 4460 | 10165 |
| GenAI | - | 1315 | 326 | 784 | 3114 | 847 | 469 | 1583 | 1003 | 693 |

**Table 1: The number of instances per dataset and identity group after identity categorization is performed.**

and found that the auto-tagging scheme had high accuracy when identifying text across all identity categories, including Christian (96%), non-Christian (98%), white (94%), non-white (94%), straight (99.8%), LGBT (99%), disability (98%), women (95%), and men (93%). Interestingly, our validation confirmed that by building from the identified Wikipedia entries, and despite this seemingly simplistic identity-tagging scheme, we were able to closely match human judgment in identifying text as related to a variety of identities.

*4.2.2 External associations.* The movie and TV datasets were associated with identity groups based on external information about the show or movie. Similarly to the use of Wikipedia to identify slurs and slang, we found Wikipedia and IMDB's crowd-sourced lists and tags related to identity and media to be usefully varied across identity and detailed in composition. Wikipedia has categories that pages can be tagged with (e.g., "Category:LGBT-related films"), and these were used to build up a list of movies and shows associated with specific identities (specific categories and URLs are given in Appx. Tables 9 and 10). In order to identify movies and shows associated with some dominant groups where such categories did not exist (e.g., TV shows about white people), larger sets of shows were collected (e.g., TV shows set in Europe) and then shows identified as not belonging to the dominant group (e.g., shows about non-white people) were removed. Additionally, user-generated tags associated with television episodes on IMDB were used to add further identity tags to episodes, using the procedures and tag lists of [46].

We note here that we marked TV and movies as related to an identity based on salience as indicated by crowd-sourcing projects as opposed to, e.g., based on the percent of actors portraying or from a specific identity. This is a purposeful choice based on an understanding from media studies of the important difference between genuine representation and casting choices. As one example, this distinction has been codified as the "Bechdel Test" wherein a movie is considered to have good representation of women if it has two named women characters who talk to each other about something other than a man [10]; we use a crowd-sourced site with a listing of movies that pass the Bechdel Test to identify movies relating to women (see Appx. Table 9). More broadly, we claim that movies and television shows individually identified by Wikipedia editors or IMDB users as related to an identity are likely to be genuinely salient for that group.[1]

*4.2.3 Identity-tagging results.* Based on all four of these identity-related tagging methodologies, counts of the number of instances associated with each general identity group per dataset resulting from this process are given in Table 1. While identity groups have a low number of resulting tagged instances for some datasets—for example, very few text instances are tagged as "straight" following the text reference identification method used for the Stormfront, TweetEval, and OpenAI datasets—when aggregated for the traditional and generative AI datasets, each identity group makes up at least approximately 1% of each dataset. The group with the lowest identified representation in the traditional dataset is straight people (0.8%) and in the generative AI dataset is white people (1.4%), while the highest represented group in both datasets is women (12% traditional, 13% generative AI). Numerically, given the large size of the datasets assembled, the representation even of the smallest group is still fairly large, with 4,460 text instances associated with straight identity marked in the traditional dataset and 326 instances associated with white identity in the generative AI dataset.

## 4.3 Automated content moderation APIs

We identified five publicly available automated content moderation APIs to audit covering a variety of types and goals. An overview of these systems, including the specific version number audited, can be found in Figure 1. Jigsaw's Perspective API is a traditional content moderation system used for moderation of online comment sections and other user-generated content. Llama Guard (accessed via Octo AI's API platform [54]) and Anthropic [3] provide specific prompts (see Appx. Figures 4, 5, and 6) to give to generative AI systems. In other words, these are large language models being used for a content moderation purpose. Anthropic states the aims of this system as focused on user-generated text (i.e., a traditional content moderation goal), and the language model used is their general

[1] In fact, instructions for Wikipedia categories make some of this explicit. For example, the Wikipedia category of "African-American Films" (https://en.wikipedia.org/wiki/Category:African-American_films) explicitly states that "This category is for Films identified by reliable sources as "African-American films". This should not be used as a catch-all for all films starring or made by African-Americans. Nor should it be used as an umbrella category for all films about African Americans."

model (Claude) [3]. Llama Guard, in contrast, focuses on generative AI conversation use cases and has fine-tuned Llama for the purpose of content moderation in this generative AI context [37]. OpenAI [48] and Google [31] have automated content moderation APIs that are separately created models aimed to be used to filter content. Google's moderation API documentation refers generally to the goal of moderating content, regardless of the source (i.e. whether AI- or user-generated). OpenAI's moderation endpoint [48] is used on the generated outputs of their AI systems (e.g., GPT-4) such that what is visible to a user of their standard web interface is the result of running the AI API followed by the moderation endpoint. This is our main use case of concern in a generative AI context; the speech that will be blocked before being shown to a user.

Each system is designed for slightly different categorization aims, from toxicity to sexual content to politics (see Fig. 1). Most systems identify a broad set of categorization aims; the only exception is Jigsaw's Perspective which is solely focused on identifying toxicity. For the remaining APIs, a few categories are consistently considered: toxicity or hate, violence, sexual content, and self-harm. Llama Guard, Anthropic, and Google additionally include categories focused on weapons, illegal drugs, and other illegal activities. Beyond these, Google has additional categories that it flags text based on that are not common to the other APIs, with scores provided on a per-category basis for the additional categories of public safety, health, religion and belief, war and conflict, politics, finance, and legal content.

Throughout, we are interested in assessing each APIs behavior from the perspective of incorrect flagging based on the dataset labels. Thus, it is also important to consider the goals of labeling for each of the datasets. All the traditional datasets were labeled with the goal of identifying toxic or hateful speech for the moderation of online user-generated comments. We note that in some cases this includes identifying violence or sexual content as toxic. The OpenAI dataset was labeled according to the OpenAI API moderation aims, i.e., a text instance is labeled as "should flag" if it contains sexual content, hate, violence, harassment, or self-harm (including subcategories such as sexual content involving minors). Finally, recall that the movie and TV datasets were labeled based on age ratings, which are themselves based on identification of violence, sexual content, inappropriate language (e.g., hateful language), and other adult themes (e.g., self-harm and illegal drug use) [6]. Thus, despite some differences in API aims, we find that the labeling and categorization aims across datasets and APIs are largely focused on identifying toxicity or hate, sexual content, and violence, and provide a reasonably consistent basis for comparison across APIs and datasets.

The types of values returned by these moderation systems also vary, including overall flags indicating content deemed violating, per-category flags indicating content identified with that category, as well as overall or per-category scores (see Fig. 1). Some systems return more than one of these types of values. While Google's returned per-category scores are meant to represent confidence that an input belongs to the associated category [31], and thus are comparable across categories, OpenAI's per-category scores must be normalized to be comparable across categories [46]. We follow the normalization mechanism described in [46]; we determine score flagging thresholds (see Appendix Table 11) per category and divide by these thresholds so that scores above 1.0 flag. While a previous audit of OpenAI determined that multiple runs per instance were useful for the stability of the results [46], we run each instance through each API only once given the size of the combined datasets and the cost of some APIs.

| API | Traditional | | GenAI | |
|---|---|---|---|---|
| | Identity | Supp. | Identity | Supp. |
| Google | non-chr. | 1.20 | non-chr. | 3.58 |
| Jigsaw | lgbt | 1.58 | men | 1.22 |
| OAI score | non-chr. | 3.26 | women | 1.88 |
| OAI flag | non-chr. | 2.19 | men | 5.05 |
| Llama | white | 2.33 | non-chr. | 2.35 |
| Anthropic | white | 1.97 | men | 2.96 |

**Table 2: Worst identity-related speech suppression values (Supp.) and associated identity group achieving that value (Identity) across content moderation APIs tested. OpenAI (OAI) has speech suppression values associated with both scores (based on $i$-medians) and flags (based on $i$-FPRs) shown, while Google and Jigsaw are both score-based and Llama Guard (Llama) and Anthropic are flag-based.**

All instances from each dataset were sent through the five automated content moderation APIs to receive categorization and/or scored results. The resulting benchmark dataset contains 543, 859 instances of traditional content moderation text and 23, 722 generative AI instances, each annotated with zero, one, or more of nine identity groups, a label indicating whether the text should be flagged as violating, and the flag and/or score results from all five content moderation APIs.

## 5 FINDINGS

Using the audit methodology described in Section 4, we collected API results across benchmark datasets. Our key findings are that:

(1) all content moderation systems tested are more likely to incorrectly suppress identity-related text, in many cases at two to three times the rate for non-identity-related text;
(2) generative AI datasets have different speech suppression patterns than traditional datasets and these are not accounted for simply by the length of the texts;
(3) speech that is incorrectly suppressed on traditional datasets is often political, while generative AI suppression is more often TV violence; and
(4) identity-related speech suppression is related to stereotypes or harms particular to an identity group, with, e.g., disability-related content more likely to be flagged for self-harm while non-Christian content flags as hateful.

In what follows we further detail these findings.

### 5.1 All APIs suffer from identity-related speech suppression

Speech suppression scores were calculated per-API across all identities and datasets. Summary results showing the identity group and associated worst speech suppression scores per-API are given in Table 2 (with per dataset results in Appx. Tables 12 and 13). We find

that all APIs suffer from identity-related speech suppression for at least one identity group, with most incorrectly flagging at least one identity's related text at 2 or 3 times the overall rate (or with median scores 2 to 3 times as high as the overall median for true negative instances). One API – Jigsaw – however, performs better, with a worst-case speech suppression score of 1.58 for LGBT-related content on the traditional data and with most identity group scores close to 1, i.e. parity. This is promising, since Jigsaw has a history of attending to this issue [14, 20, 24], demonstrating that steps can be taken by other APIs to reduce identity-related speech suppression. While speech suppression scores are useful for comparing across APIs, it's useful to understand these scores in the context of the underlying false positive rates; 0.2 and 0.03 for OpenAI, 0.2 and 0.06 for Llama Guard, and 0.6 and 0.08 for Anthropic on the traditional and generative AI data, respectively. For Anthropic this results in an $i$-FPR for men on the generative AI data of 0.247; in other words, about 25% of content about men in the generative AI dataset that has a label indicating that it should not be flagged as violating is filtered out.

Considering the trends across all identity groups and APIs (see Fig. 2 and Appx. Tables 14 and 15), we find that while most identity groups suffer from speech suppression under some dataset and content moderation system, there is consistently high speech suppression related to non-Christian religions across APIs. Additionally, many APIs suffer from speech suppression on both dominant and marginalized groups. For example, Llama Guard has high speech suppression rates for both white and non-white content on the traditional dataset. While there are variations by specific dataset within the traditional and generative AI categories (Appx. Tables 12 and 13), the overall trends hold.

*5.1.1 Regression analysis.* To further validate the key finding that these APIs suppress identity-related speech, we conducted a regression analysis. Consistent with our speech suppression definitions and approach, we evaluate each API by focusing on cases where they suggest censoring content that should not actually be censored. For the three APIs that return binary flags (OpenAI, Anthropic, and Llama Guard) we train logistic regressions to predict whether the flag on a given true negative labeled text instance results in a false positive (i.e., is incorrectly flagged). For the three APIs that return continuous scores between 0 and 1 (OpenAI, Jigsaw, and Google), we train linear regressions that predict higher scores, restricting the data to true negatives—content that, according to ground truth, *is* appropriate and should not be censored, and where higher assigned scores are more incorrect. In all six models, inputs include our nine identity tags (binary values indicating whether the content pertains to attributes like disability, LGBT, Christians, or women), as well as three other descriptive attributes: whether the text was identity-tagged because it contains a slur or slang term, whether it comes from a generative AI-focused dataset, and its word count (centered at 0 and scaled to unit variance).

Across all regression models (see Table 3) the immediately apparent result, confirming the above analysis, is that identity-related content is statistically significantly likely to be incorrectly flagged or incorrectly scored higher than non-identity-related text. This is true for most APIs, whether the identity tag is for a marginalized group (women, non-Christian, LGBT, non-white, and disability) or for most dominant groups (men, white). Breaking from this trend, however, most of the APIs do not display significant speech suppression towards straight content, and some are even less likely to suppress straight-related speech than other speech. All other identity groups have one or two APIs which were not found to suppress related speech, with the exception of non-Christian content; consistent with the literature on the difficulty in removing violent associations with Muslim content [1], non-Christian content was found to be speech suppressed across all APIs, including some with notably high effect sizes.

## 5.2 There are differences in identity-related speech suppression for traditional and generative AI datasets

We find that the differences between the generative AI and traditional datasets matter for identity-related speech suppression. Some content moderation systems generally have less identity-related speech suppression on some identity groups in the traditional data while others have less suppression for some identity groups on the generative AI data. For example, Google suffers from more non-Christian speech suppression on the generative AI data than the traditional data (see Fig. 2). Surprisingly, this does not directly align with the goals of these systems; Jigsaw's Perspective API was designed for toxicity-detection for a traditional content moderation task, yet suffers from more identity-related speech suppression for the LGBT group on the traditional data (although these differences are small given that this model does comparatively well overall). If we compare identity groups within demographic categories (race, gender, and so on, see Fig. 2 and Appendix Tables 14 and 15), we find that on the traditional content moderation datasets, with the exception of race, the marginalized group has worse speech suppression across APIs. The trend is less clear for the generative AI related data. For example, across all APIs on the generative AI dataset white and non-white groups experience similar levels of speech suppression as each other, and the trend is mixed for sexual orientation.

*5.2.1 Regression analysis.* In order to further test this finding, we included three key variables as part of the regression analysis (described in Section 5.1.1): a Boolean flag indicating whether the instance was text from the traditional or generative AI dataset, a flag indicating whether the instance was identity tagged based on containing a slur or slang term, and a (scaled) count of number of words per instance. Increased word length was found to make almost all APIs (with the exception of Llama Guard) more likely to incorrectly flag. Using the generative AI dataset Boolean flag, we found that generative AI-focused text was less likely to be incorrectly speech suppressed, with large effect sizes, despite that the average word count of the instances in the generative AI datasets is larger. One possible interpretation is that despite the difficulty of correctly flagging longer text instances, the specific datasets contained in the generative AI set were easier content moderation problems. This may especially have been true for the TV and Movies datasets; given that the content comes from Wikipedia, IMDB, and TMDB, we expect that none of it would be particularly inappropriate, and they may contain fewer hard-to-classify instances than the traditional datasets.